Abstract

The goal of this research project was to discern whether Sturmfels' tropical Grassmannian possesses a natural algebraic characterization similar to that found in Hodge and Pedoe 11.3.1-2. Goodman and Wallach describe the process by which the classical Grassmannian is attained via cosets of $GL(n)$. In the tropical case, our general linear group is precisely the group of monomial matrices discussed by Izhakian [6]. Our goal is an equivariant characterization of $G_{tr}(2,4)$, and in pursuit of this, we make use of a tropical matrix group that serves as an analogue of the general linear group and define tropical linear subspaces. We show that tropical linear subspaces, defined as vector spans, are not an orbit under the group action of monomial matrices, but that the action allows us to decompose $G_{tr}(2,n)$ into what we have termed "inequality classes."

An Equivariant Characterization of the Tropical Grassmannian $\mathrm{G}_{\mathbf{tr}}(2,4)$

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1 Introduction

This project began in Spring 2017 with the goal of defining an equivariant characterization of the tropical Grassmannian $G_{tr}(2,4)$ introduced by Speyer and Sturmfels in"Tropical Mathematics" [5]. The classical Grassmannian is a means of geometrically parametrising every subspace of a linear space. This object is well understood and has both polynomial and equivariant characterizations. Sturmfels and Speyer introduce their tropical Grassmannian as the intersection of hyperplanes defined by Grassmann-Plücker relations. However, this equational representation is cumbersome to work with even in low dimensional cases. This set us in search of an equivariant characterization of the tropical Grassmannian in terms of cosets of some tropical general linear group. By mid spring, we independently discovered Izhakian's group of monomial matrices (our candidate for a tropicalized general linear group), and had experimented with a tropical analogue of classical linear algebra. Early summer was largely spent researching the classical Grassmannian and plotting a plan of attack for a tropical equivariant characterization. We closed our research by telescoping in on polynomial and linear span characterizations of tropical lines and planes. In classical linear algebra these characterizations are equivalent, but in tropical linear space, they produce completely different results. In particular, planes defined by polynomials are not closed under scalar multiplication, so we opted for a vector span characterization. Though we were initially disappointed to learn that the orbits of our subspaces are not all of $G_{tr}(2,4)$, we analyzed the orbit structure of elements of $G_{tr}(2,4)$ under the group action of monomial matrices and provide a formula for enumerating the inequality classes of $G_{tr}(2,n)$. Funding for this project courtesy of the Laureates Program at Hillsdale College.

2 The Free Tropical Semimodule $\mathbb{R}^n_{max}$

Definition. For $a, b \in \mathbb{R}$, their **tropical sum** is $a \oplus b = \max(a, b)$ and their **tropical product** is $a \odot b = a + b$ [5].

With tropical addition and multiplication defined, we now turn our attention to Kohonen and Tohge's description of of the free tropical semimodule [7].

Definition. The tropical semimodule $\mathbb{R}^n_{max}$ is precisely the set of n-tuples with entries from $\mathbb{R} \cup \{-\infty\}$ equipped with coordinate-wise tropical addition and scalar multiplication. That is, for $u, v \in \mathbb{R}^n_{max}$ and $\lambda \in \mathbb{R} \cup \{-\infty\}$:

(i) $u \oplus v = (u_1 \oplus v_1, \ldots, u_n \oplus v_n) = (\max(u_1, v_1), \ldots, \max(u_n, v_n))$;

(ii) $\lambda = (\lambda \odot u_1, \ldots, \lambda \odot u_n) = (\lambda + u_1, \ldots, \lambda + u_n)$.

It should be clear that our additive identity is the tuple whose entries are all negative infinity. A little picture drawing of the 2-dimensional case will reveal that each unique tuple up to scalar multiples corresponds to a line parallel to the line $x = y$. Though these elements are not actually vectors, as they do not form a vector space, they do resemble vectors in many respects, to be seen later. We will frequently refer to elements of the free tropical semimodule as **tropical vectors**.

3 The Group of Monomial Matrices

Group Theory

Recall that a **group** is a set G equipped with an operation $(*)$ such that $\forall$ $a, b, c \in G$:

(i) **Closure** $a * b \in G$;

(ii) **Associativity** $a * (b * c) = (a * b) * c$;

(iii) **Identity** $\exists\ \mathbb{I} \in G$ such that $\forall a \in G$, $a * \mathbb{I} = \mathbb{I} * a = a$;

(iv) **Inverses** $\forall a \in G$, $\exists\ a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = \mathbb{I}$.

A **subgroup** S of a group G is a subset of the elements of G which satisfy the group axioms. We denote "H is a subgroup of G" by $H < G$.

3.1 The Group of Monomial matrices

We are able to construct groups of matrices, that is, sets of matrices equipped with matrix multiplication that satisfy all group requirements. One such example is the general linear group $GL(n)$ which consists of all invertible $n \times n$ matrices. The tropical analogue of this group is precisely the group of monomial matrices.

Definition. The set of **monomial matrices** is the set of square matrices with entries from $\mathbb{R} \cup \{-\infty\}$ such that each row and collumn has exactly one finite entry.

The products of these matrices with vectors and other monomial matrices is the same as the classical cases with the standard arithmetic swapped for tropical arithmetic.

Proposition. *Monomial matrices form the group $GL_{tr}(n)$ under tropical matrix multiplication.*

Proof. To prove that $GL_{tr}(n)$ is a group under tropical matrix multiplication, we need only confirm that for arbitrary matrices $A, B \in GL_{tr}(n)$, each group axiom is satisfied.

(i) **Closure**

For matrices $A, B \in GL_{tr}(n)$, $AB = m_{ij} = a_{i,1} \odot b_{1,j} \oplus \cdots \oplus a_{i,j} \odot b_{j,i} \oplus \cdots \oplus a_{i,n} \odot b_{n,j}$. For fixed i, $\exists!$ k such that $a_{i,k} \neq -\infty$. Similarly, for fixed j $\exists!$ k such that $b_{k,j} \neq -\infty$. It is clear then that for fixed i and j, $(m_{i,j})$ $\neq -\infty$ if and only if $\exists$ k such that $a_{i,k}, b_{k,j} \neq -\infty$. Then for fixed i, $\exists!$ j such that $(m_{i,j}) \neq -\infty$. Similarly for fixed j. Therefore, the product of monomial matrices is a monomial matrix.

(ii) **Associativity**

Associativity comes free by virtue of the associativity of tropical arithmetic.

(iii) **Identity**

The identity in $GL_{tr}(n)$ is

$$\mathbb{I} = \begin{pmatrix} 0 & -\infty & \cdots & -\infty \\ -\infty & 0 & \cdots & -\infty \\ \vdots & \vdots & \ddots & \vdots \\ -\infty & -\infty & \cdots & 0 \end{pmatrix}.$$

It is immediately clear by inspection that for $A \in GL_{tr}(n)$, $\mathbb{I}A = A\mathbb{I} = A$.

(iv) **Inverses** For $A \in GL_{tr}(n)$ where $A = (a_{ij})$, it is clear by the definition of a monomial matrix that for fixed i, $\exists!$ j such that $a_{i,j} \neq -\infty$. Then for a matrix $A^{-1} = (x_{j,i})$ where $x_{j,i} = -a_{i,j}$, we have

$$[A^{-1}A]_{ij} = [AA^{-1}]_{ij} = \begin{cases} a_{i,j} + (-a_{i,j}) = -a_{i,j} + a_{i,j} = 0 & i = j \\ -\infty & i \neq j \end{cases}$$

Thus, $AA^{-1} = \mathbb{I}$.

Therefore, the set of monomial matrices is a group under tropical matrix multiplication. □

With a tropical matrix group in hand, we need only define a few more group theory related objects before considering the classical and tropical Grassmannians.

3.2 Group Actions on a set

Definition. A group G **acts on a set** X if there is a homomorphism that sends an element of G to a permutation of the elements of X (S_X).

In the particular case of groups of matrices acting on linear spaces, our matrices map vectors to vectors through matrix-vector multiplication.

Definition. For a group G acting on a set X and x $\in X$, the **stabilizer** of x with respect to G is $\text{stab}_G(x) = \{\text{g} \in G : \text{g(x)} = \text{x}\}$.

For a group G acting on a set X, the stabilizer of x $\in$ X is in fact a subgroup of G. [1].

Definition. For an element x in a set X and a group G, the **orbit** of x is precisely $\{\text{y} \in X\text{: } \exists \text{ g} \in G \text{ such that g(x)} = \text{y}\}$.

Definition. For a subgroup H of a group G, the **quotient** G/H is $gH = \{gh : g \in G, h \in H\}$.

A useful way of thinking about quotients of groups is that they impose an **equivalence relation** among the elements of G. For $H < G$, we can think of H acting on G via the group operation where gH is simply the orbit of g by H. This overview of basic group theory is sufficient for equivariant constructions of the Grassmannian.

4 Grassmannians

The Grassmannian $G(d, n)$ is the set of d-dimensional subspaces of n-dimensional space. There are multiple equivalent ways of characterizing the classical Grassmannian. Of these, we will focus upon two: the equational and the equivariant.

4.1 Wedge Products

To introduce the classical equational characterization, we first need a little multilinear algebra courtesy of Goodman and Wallach Appendix B.2.4 [4].

Definition. For $u, v \in \mathbb{R}^n$, the **wedge product** u maps ordered pairs of vectors to elements of $\bigwedge^2 \mathbb{R}^n$ such that

(i) $u = 0$;

(ii) $u = -v$.

A useful way of thinking about wedge products is that they are used to construct **multivectors**. A simple illustration might be that a scalar can be thought of as describing only magnitude, a vector possesses magnitude and direction, and

the wedge product of two vectors u possess magnitude and two directions. This wedge product has a natural characterization as the $n \times 2$ matrix

$$u \cong \begin{pmatrix} \vdots & \vdots \\ u & v \\ \vdots & \vdots \end{pmatrix}.$$

It's clear that given a vector space V of dimension n, we have $\dim(\bigwedge^k V) = \binom{n}{k}$, which becomes clear when one realizes that the basis vectors of $\bigwedge^k V$ are precisely each possible k-fold wedge product of the basis vectors of V.

4.2 Equational Characterization of the Classical Grassmannian

Definition. The **projective space** $\mathbb{P}(V)$ of a vector space V is the set of one-dimensional subspaces of V. That is, for $x \in V \backslash \{0\}$, $[x] \in \mathbb{P}(V)$ is the line through x and the origin [4].

Definition. An **algebraic variety** is the set of solutions to a set of polynomial equations over the real or complex numbers.

For an n-dimensional vector space V, and a k-dimensional subspace $\Lambda \subset V$ spanned by the vectors $v_1 \dots v_k$, we can associate Λ with the multi-vector $\lambda = v_1 \wedge v_2 \wedge \cdots \wedge v_k \in \bigwedge^k(V)$ and then a point in $\mathbb{P}(\bigwedge^k V)$ via the Plücker embedding

$$\Psi : \text{G(k,n)} \to \mathbb{P}(\bigwedge^k V)$$

The Plücker embedding simply sends k-dimensional subspaces of V to 1-dimensional subspaces of $\bigwedge^k V$, and then sends these 1-dimensional subspaces to unique points in $\mathbb{P}(\bigwedge^k V)$. The points associated with Λ may be found by arranging its basis vectors of into a $n \times k$ matrix and then computing its maximal minors (the determinants of the $k \times k$ submatrices) [2]. These coordinates in $\mathbb{P}(\bigwedge^k V)$ are the **Plücker coordinates**, the homogeneous coordinates of $\Phi(\Lambda)$ in $\mathbb{P}(\bigwedge^k V)$. Each collection of such coordinates $[x]$ uniquely corresponds to the subspace Λ from which said coordinates originated from [3]. An additional property of these coordinates, deriving from the independence of the vectors in the span of Λ, is that they are solutions to $\binom{n}{d}$-term polynomials. Thus, the Grassmannian is an algebraic variety characterized by the vanishing of a general $\binom{n}{d}$-term polynomial.
The simplest example of these general polynomials is found in the case where $n = 4$ and $d = 2$.

Example. Consider a subspace $\Lambda \subset \mathbb{R}^4$ where $\dim(\Lambda) = 2$. This subspace is spanned by two linearly independent vectors u and v. As mentioned earlier, we

can arrange these vectors into a 4×2 matrix

$$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \\ u_3 & v_3 \\ u_4 & v_4 \end{pmatrix}$$

The maximal minors (the determinants of the 2×2 submatrices) are our Plücker coordinates. These coordinates may be indexed by which rows were used to construct their corresponding submatrices. For instance, the coordinate $X_{12} = \det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$. Each unique set of Plücker coordinates corresponding to a subspace Λ is a unique solution to the general polynomial

$$X_{12}X_{34} + X_{13}X_{24} + X_{14}X_{23} = 0$$

For those curious as to why this is the case for all subspaces Λ, further reading may be found in Harris [2].

4.3 Equivariant Characterization of the Classical Grassmannian

Of more immediate interest is the equivariant characterization of the classical Grassmannian.

Theorem. *Let V be a vector space with* $\dim(V) = n$ *such that $GL(n)$ acts on V and $orb(x) = V$ for all $x \in V$. Then for any linear subspace $S \subset V$ with* $\dim(S) = d$*, the Grassmannian $G(d, n) \cong GL(n)/stab(S)$.*

Proof. Let V be a vector space with $\dim(V) = \mathrm{n}$ and let S have $\dim(S) = \mathrm{d}$. Now, S is expressible as $\mathrm{span}(v_1, \ldots, v_d)$ where $v_i \in V$ and $\forall i \neq j$, $v_i \neq v_j$. Now, $\forall$ A $\in$ GL(n) such that the matrix-vector product A(v_i)= $\lambda_i v_i$ $\forall$ $v_i \in \{v_1, v_2, \ldots, v_d\}$, A $\in$ stab(S) because $\mathrm{span}(\lambda_1 v_1, \ldots, \lambda_d v_d) = \mathrm{span}(v_1, v_2, \ldots, v_d)$. Then for $M \in$ GL(n) and $h \in$ stab(S), the quotient GL(n)/stab(S) = {M(h)}. Note that each coset in GL(n)/stab(S) is an equivalence class of matrices and that non-identity cosets are equivalence classes of matrices that map S to a given subspace $T \neq S$. This means that every non-trivial coset $K \in$ GL(n)/stab(S) sends S to some subspace T where $S \neq T$. Therefore, there is a one-to-one correspondance between d-dimensional subspaces of V and the elements of GL(n)/stab(S). □

This may perhaps be more clearly demonstrated through a concrete example.

Example. Consider the subspace $S = \mathrm{span}((1,0,0,0),(0,1,0,0)) \subset \mathbb{R}^4$. It is clear that every matrix

$$M = \begin{pmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix}, \ a, b, c, d, e, f, g \in \mathbb{R}$$

preserves S. Thus, each such matrix is an element of stab(S). Then non-identity elements of the quotient group GL(4)/stab(S) are equivalence classes of matrices that map S to a subspace $T \neq S$, $\dim(T) = \dim(S) = 2$. By construction of the quotient group, it is impossible for two distinct cosets to map S to the same subspace. Thus, there is a one-to-one correspondence between G(2,4) and GL(4)/stab(S).

4.4 Tropical

Sturmfels and Speyer establish a definition of the tropical Grassmannian for $\binom{n}{d}$-dimensional space [5]. We consider the $\binom{n}{d}$-dimensional space $\mathbb{R}^{\binom{n}{d}}$ whose coordinates $\mathrm{X}_{i_1 \ldots i_d}$ we index by d-element subsets $\{i_1, \ldots i_j\} \subset \{1, 2, \ldots, n\}$. Let S be any (d-2)-element subset of $\{1, 2, \ldots, n\}$ and let i, j, k, l be any four distinct indices in $\{1, 2, \ldots, n\} \backslash S$. We can then describe the **three-term Grassmann Plücker relation** via the general polynomial

$$p_{S,i,j,k,l} = X_{Sij} \odot X_{Skl} \oplus X_{Sik} \odot X_{Sjl} \oplus X_{Sil} \odot X_{Sjk}.$$

From here, Speyer and Sturmfels define the **three-term Tropical Grassmannian** to be the intersection

$$G_{tr} = \bigcap_{S,i,j,k,l} \mathcal{H}(p_{S,i,j,k,l}) \subset \mathbb{R}^{\binom{n}{d}}$$

This characterization of the Tropical Grassmannian closely models the classical equational characterization. However, it is quite cumbersome to compute, raising question of whether more concise characterizations exist. In the classical case, the equational and equivariant versions of the Grassmannian are equivalent. If this hold true in the tropical case, a more elegant characterization of the tropical Grassmannian might be possible with algebra.

5 Tropical Subspaces as spans of tropical vectors

It might initially occur to the reader to define Tropical linear subspaces as the zero loci of a family of polynomials, as may be done in the classical setting. A definition of this sort, however, is not closed under scalar multiplication.

Consider the solution set of the tropical polynomial

$$\mathbb{K} = \{(x, y, z) \mid -4 \odot x \oplus -10 \odot y \oplus -2 \odot z = 0\}.$$

The result is the surface of a sort of half cube with a corner at (4, 10, 2) in $\mathbb{R}^3_{\max}$. For any $\lambda \neq 0$, $\lambda(4, 10, 2) = (\lambda \odot 4, \lambda \odot 10, \lambda \odot 2) = (\lambda + 4, \lambda + 10, \lambda + 2) \notin \mathbb{K}$. It is clear then that $\lambda\mathbb{K} \neq \mathbb{K}$, that is, this would-be linear subspace is not closed under scalar multiplication.

Clearly, polynomials are not the way forward. We can, however, preserve all the nice properties of classical linear subspaces by defining the tropical case in terms of linear combinations of tropical vectors.

Definition. A one-dimensional tropical linear subspace (a tropical line) in $\mathbb{R}^n_{max}$ is the equivalence class of all tropical scalar multiples of a tropical vector v.

Definition. For a set of tropical vectors $V = \{v_1, v_2, \ldots, v_k\}$ such that

$$v_i = (v_{i_1}, v_{i_2}, \ldots, v_{i_n}),$$

$$\text{span}(v_1, v_2, \ldots, v_k) = (\lambda_1 v_{1_1} \oplus \lambda_2 v_{2_1} \oplus \cdots \oplus \lambda_k v_{k_1}, \ldots, \lambda_1 v_{1_n} \oplus \lambda_2 v_{2_n} \oplus \cdots \oplus \lambda_k v_{k_n}).$$

The following lemma will be useful going forward.

Lemma. *Any span of d vectors in $\mathbb{R}^n_{max}$ is entirely determined by their region in the $(n-1)$-dimensional hyperplane at $x_n = 0$.*

Proof. Let $S = \text{span}(v_1, v_2, \ldots v_d)$. It follows from the definition that $\text{span}(v_1, v_2, \ldots v_d)$ $= \text{span}(\lambda_1 \odot v_1, \lambda_2 \odot v_2, \ldots, \lambda_d \odot v_d)$ for all $\lambda_i \neq -\infty$. Thus, for a vector v_i $= (v_{i_1}, v_{i_2}, \ldots, v_{i_n})$, we may choose $\lambda_i = -v_{i_n}$. Then we can express S as the span of vectors $\lambda_i \odot v_i$ each with n^{th} coordinate value 0. The result will be a collection of points in the (n-1)-dimensional hyperplane at $x_n = 0$. □

With this in mind, we will proceed assuming that the nth coordinate of each n-dimensional vector (together forming a d-dimensional span) is 0. In the case of $\mathbb{R}^3_{max}$, we will assume that our vectors all have $z = 0$.

Definition. The **delta vector** corresponding to the vectors u and v is the n-tuple

$$\Delta^{u,v} = (v_1 - u_1, \ldots, v_n - u_n).$$

Theorem. *Let u and v be n-dimensional tropical vectors with delta vector $\Delta^{u,v}$ and let $M \in G^n_{tr}$, then $\Delta^{Mu,Mv}$ is a permutation of $\Delta^{u,v}$.*

Proof. Fix $i \in \{1, \ldots, n\}$. By definition of a monomial matrix, there is a unique $j \in \{1, \ldots, n\}$ such that $M_{ij} \neq -\infty$. Then the ith entry of the vector Mu is $u_j \odot M_{ij}$. Similarly, the ith entry of Mv is $v_j \odot M_{ij}$. So then

$$\Delta_i^{Mu,Mv} = v_j + M_{ij} - (u_j + M_{ij}) = v_j - u_j$$

So each $\Delta_i^{Mu,Mv}$ is equal to a unique $\Delta_j^{u,v}$. Hence $\Delta^{Mu,Mv} = \sigma(\Delta^{u,v})$ where $\sigma \in S_n$. □

While the spans of tropical vectors are closed under tropical scalar multiplication and tropical vector summation, a given span of vectors may be properly contained in another span of vectors.

Definition. Let $\{u_i\}$ and $\{v_i\}$ be sets of k linearly independent n-dimensional tropical vectors. We call $\text{span}(v_i)$ an **extension** of $\text{span}(u_i)$ if $\text{span}(u_i) \subset \text{span}(v_i)$.

Definition. Let $u, v, u^*, v^* \in \mathbb{R}^n_{max}$, we say that $\text{span}(u^*, v^*)$ is the **maximal extension** of $\text{span}(u, v)$ if it contains every extension of $\text{span}(u, v)$.

With this in hand, we can finally offer our definition of a tropical linear subspace.

Definition. A d-dimensional tropical subspace of $\mathbb{R}^n_{max}$ is the equivalence class of spans of vectors that have the same maximal extension.

We motivate this definition with the following example.

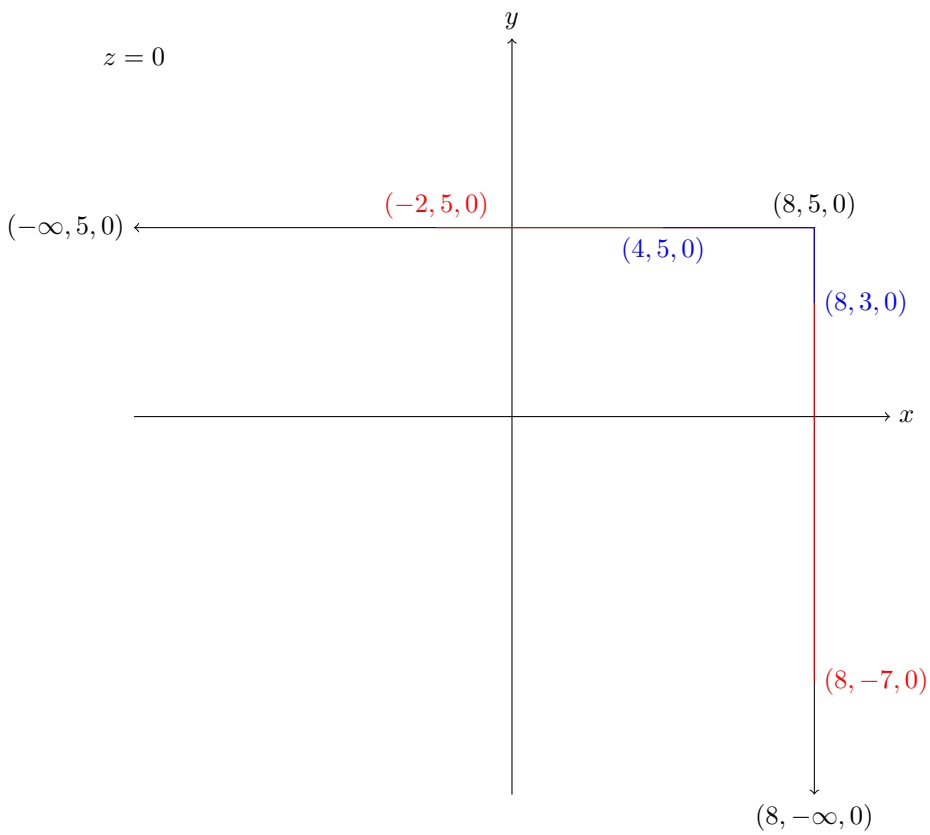
Example. Let $u_1 = (4, 5, 0)$, $v_1 = (8, 3, 0)$, $u_2 = (-2, 5, 0)$, and $v_2 = (8, -7, 0)$. $\text{span}(u_1, v_1)$ gives us a plane passing through a region in z = 0. We write out this vector span as follows:

$$\text{span}(u_1, v_1) = \Big((a \odot 4) \oplus (b \odot 8),\ (a \odot 5) \oplus (b \odot 3),\ a \oplus b \Big)$$

Because we want $a \oplus b = 0$, it is clear that a or b must equal zero. So if $a \neq 0$, then $a < 0$ and $b = 0$. Similarly, if $b \neq 0$ then $b < 0$ and $a = 0$. To plot $\text{span}(u_1, v_1)$, we perform the following calculations:

1. If $a = 0$, $b \leq 0$, then $\forall b \leq 0$, $y = 5$ and $4 \leq x \leq 8$
2. If $a \leq 0$, $b = 0$, then $\forall a \leq 0$, $x = 8$ and $3 \leq y \leq 5$.
3. When $a = b = 0$, we get the point (8, 5, 0).

A second round of calculation will show that the region at z = 0 in $\text{span}(u_1, v_1)$ is a proper subset of $\text{span}(u_2, v_2)$, so $\text{span}(u_2, v_2)$ is an extension of $\text{span}(u_1, v_1)$ and thus, are each representatives of the same tropical 2-space. Their maximal extension is the span of the vectors $u^* = (-\infty, 5, 0)$ and $v^* = (8, -\infty, 0)$. The intersection of each vector span with $z = 0$ is pictured below.



Since two different spans of vectors can produce the same tropical linear space, it is perfectly natural to create a collection of maps that take spans of vectors in a given tropical space and output different spans within the same tropical space. So if α, β are two different spans in a tropical subspace $S \subset \mathbb{R}^n_{max}$, then denote by $\Phi_{\alpha,\beta}$ the map that sends α to β. This mapping is useful because, as we saw earlier, the action of a monomial matrix on the span of vectors u, v cannot change the relative distances between their coordinates. This mapping allows us to choose a span with a desirable delta vector from which we can map to another vector span with a permutation of said delta vector.

Orbits of Tropical Linear Spaces

We have defined our linear subspaces and our group, now we can focus on $G_{tr}(2, 4)$. As discussed in the previous section, we can associate to a maximal extension to each tropical linear subspace since the maximal extensions are, by definition, unique to a given subspace. Therefore, we can learn what structure $G_{tr}(2, 4)$ by studying its maximal extensions. We note that these maximal extensions are determined by the relative inequalities between the coordinates

of the spanned vectors. This motivates the creation of the following entity:

Definition. An inequality class n-tuple $\mathcal{C}$ is an n-tuple with entries from the collection $\{=, <, >, \underset{-\infty}{<}, \underset{-\infty}{>}\}$ that obey the following rules:

1. At least one entry is $\underset{-\infty}{<}$
2. At least one entry is $=$
3. Permutations are equivalent. For example:

$$(\underset{-\infty}{<}, <, >, >, =) \sim (\underset{-\infty}{<}, >, <, >, =)$$

4. If a tuple has at least one pair $\underset{-\infty}{<}, \underset{-\infty}{>}$, then it is equivalent to itself with each $<, >, \underset{-\infty}{<}$, and $\underset{-\infty}{>}$ flipped. For example:

$$(\underset{-\infty}{<}, \underset{-\infty}{>}, <, <, =) \sim (\underset{-\infty}{<}, \underset{-\infty}{>}, >, >, =)$$

$$(\underset{-\infty}{<}, \underset{-\infty}{>}, \underset{-\infty}{>}, =) \sim (\underset{-\infty}{<}, \underset{-\infty}{>}, \underset{-\infty}{<}, =)$$

This eliminates double counting of certain inequality classes where the order of the vectors in the span doesn't actually matter.

Inequality class tuples give us a way to encode the behavior of vectors composing a subspace. That is, given two vectors u, v which span a tropical linear space, one can create a corresponding inequality class tuple $\mathcal{C}^{u,v}$ and use it to create the maximal extension of $\text{span}(u, v)$. We first take a look at $\Delta^{u,v}$ and define

$$\text{pos}(\Delta^{u,v}) = \{\Delta_i^{u,v} \mid \Delta_i^{u,v} > 0\}$$

$$\text{neg}(\Delta^{u,v}) = \{\Delta_i^{u,v} \mid \Delta_i^{u,v} < 0\}$$

We construct $\mathcal{C}^{u,v}$ as follows:

1. If $\Delta_i^{u,v} = 0$, then $C_i^{u,v}$ is $=$.
2. If $\Delta_i^{u,v} = \max(\text{pos}(\Delta^{u,v}))$, then $\mathcal{C}_i^{u,v}$ is $\underset{-\infty}{<}$.
3. If $\Delta_i^{u,v} = \inf(\text{neg}(\Delta^{u,v}))$, then $\mathcal{C}_i^{u,v}$ is $\underset{-\infty}{>}$.
4. If $\Delta_i^{u,v} \in \text{pos}(\Delta^{u,v}) - \max(\text{pos}(\Delta^{u,v}))$, then $\mathcal{C}_i^{u,v}$ is $<$.
5. If $\Delta_i^{u,v} \in \text{neg}(\Delta^{u,v}) - \inf(\text{neg}(\Delta^{u,v}))$, then $\mathcal{C}_i^{u,v}$ is $>$.

Now that we have the corresponding $\mathcal{C}^{u,v}$, we can use it to construct u^* and v^* as follows:

1. If $\mathcal{C}_i^{u,v}$ is $\underset{-\infty}{<}$, then $u_i^* = -\infty$.
2. If $\mathcal{C}_i^{u,v}$ is $\underset{-\infty}{>}$, then $v_i^* = -\infty$.
3. Else, $u_i = u_i^*$ and $v_i = v_i^*$.

Definition. We say that $\operatorname{span}(u_1, v_1)$ and $\operatorname{span}(u_2, v_2)$ have the same inequality class if their inequality tuples are identical.

Theorem. *Let $n \geq 4$. If n is even, then the number of inequality classes of 2-dimensional subspaces of n-dimensional space is*

$$\binom{n+2}{n-2} - \sum_{m=0}^{\frac{n-4}{2}} \left(\sum_{k=0}^{n-4-2m} \left(\sum_{g=0}^{\lfloor \frac{n-3-2m-k}{2} \rfloor} \left\lfloor \frac{n-2-2m-k-g}{2} \right\rfloor \right) \right).$$

If n is odd, then the number of inequality class tuples is

$$\binom{n+2}{n-2} - \sum_{m=0}^{\frac{(n-1)-4}{2}} \left(\sum_{k=0}^{n-4-2m} \left(\sum_{g=0}^{\lfloor \frac{n-3-2m-k}{2} \rfloor} \left\lfloor \frac{n-2-2m-k-g}{2} \right\rfloor \right) \right).$$

Proof. Inequality class tuples are constructed from an alphabet of five symbols; the first and final entries of each tuple are fixed. Therefore, there are $n-2$ positions to be filled by our alphabet of 5 symbols. Symbol repetition is permitted and symbol order does not matter, so the number of strings up to permutation is

$$\left(\binom{5}{n-2} \right) = \binom{n+2}{n-2}.$$

We recall part four of the definition of an inequality class tuple: if an inequality class tuple has an equal quantity of $\underset{-\infty}{<}$ and $\underset{-\infty}{>}$ symbols, then it is equivalent to itself with each $<$ replaced with $>$ and vice versa. Given such an n-tuple, let us assume n is even. Since there is at least one $\underset{-\infty}{<}, \underset{-\infty}{>}$ pair, we need only consider what occupies the remaining $n-3$ positions. We count the number of possible $\underset{-\infty}{<}, \underset{-\infty}{>}$ pairs past one with m such that $0 \leq m \leq \frac{n-4}{2}$. Fixing m, there are $n-3-2m$ positions that can be occupied by symbols that are not $\underset{-\infty}{<}$ or $\underset{-\infty}{>}$. For fixed $0 \leq k \leq n-3-2m$, we will say there are k occupied by $=$. In the remaining $n-3-2m-k$ positions, we now specify the quantity of $>$ symbols, g with the requirement that $0 \leq g \leq \lfloor \frac{n-3-2m-k}{2} \rfloor$. Now, of the remaining $n-3-2m-k-g$ spots, we choose from an alphabet of two symbols, $<$ and $\underset{-\infty}{<}$, giving us

$$\left(\binom{2}{n-3-2m-k-g} \right) = \binom{n-2-2m-k-g}{n-3-2m-k-g} = n-2-2m-k-g$$

arrangements. Of these, there are

$$\left\lfloor \frac{n-2-2m-k-g}{2} \right\rfloor$$

which are double counted by the binomial. Letting m, k, and g vary, there are

$$\sum_{m=0}^{\frac{n-4}{2}} \left(\sum_{k=0}^{n-4-2m} \left(\sum_{g=0}^{\lfloor \frac{n-3-2m-k-1}{2} \rfloor} \left\lfloor \frac{n-2-2m-k-g}{2} \right\rfloor \right) \right)$$

total inequality class tuples that are double counted by $\binom{n+2}{n-2}$, making the number of distinct inequality class tuples equal to that expressed in the theorem statement.

Now supposing n is odd, we again let m be one less than the quantity of $\underset{-\infty}{<}, \underset{-\infty}{>}$ pairs. We note that $n-3$ is even, which means it is possible to fill every position before last in the tuple with $\underset{-\infty}{<}, \underset{-\infty}{>}$ pairs. Since the tuple consisting of $\underset{-\infty}{<}, \underset{-\infty}{>}$ pairs and an equal sign is not double counted by $\binom{n+2}{n-2}$, we require $0 \leq m \leq \frac{n-5}{2}$. Everything else remains as in the case where n is even, giving us

$$\sum_{m=0}^{\frac{n-5}{2}} \left(\sum_{k=0}^{n-4-2m} \left(\sum_{g=0}^{\lfloor \frac{n-3-2m-k}{2} \rfloor} \left\lfloor \frac{n-2-2m-k-g}{2} \right\rfloor \right) \right)$$

double counted inequality class tuples. The simple observation that $n-5 = (n-1)-4$ completes the proof. □

The importance of everything developed over the course of this section is revealed in the following theorem and its corollary:

Theorem (**Inequality Class Preservation**). *Given a two-dimensional linear subspace $S \subset \mathbb{R}^n_{tr}$, for all $M \in GL_{tr}(n)$, S and $(M)S$ have the same inequality class.*

Proof. We know by the theorem in the previous section that given two space spanning vectors u and v, $\sigma(\Delta^{u,v}) = \Delta^{Mu,Mv}$ for some $\sigma \in S_n$. Then the construction of $C^{u,v}$ and $C^{Mu,Mv}$ differs only by σ, which means they are the same inequality tuple by definition. Thus, $\text{span}(u,v)$ and $\text{span}(Mu,Mv)$ have the same inequality class by definition. □

Corollary. *$G_{tr}(2,n)$ is not an orbit of $GL_{tr}(n)$.*

This guarantees that there is no simple characterization of the form $G_{tr}(2,n)$ in terms of cosets of $GL_{tr}(n)$. This does not, however, mean that there is no useful information that can be given by the action of a monomial matrix on an inequality class.

Definition. An inequality class of 2-dimensional subspaces of $\mathbb{R}^n_{tr}$ is well-behaved if it is an orbit under the group action by $GL_{tr}(n)$.

Given some n, how many inequality classes of two dimensional subspaces of n dimensional space are well-behaved?

Future Work: Classification of Elements of $G_{tr}(2,4)$

The theorem tells us that there are 13 distinct inequality classes whose members compose $G_{tr}(2,4)$. Are any of these not well-behaved?

Example. Consider the span of the vectors

$$u = (0, 5, 1, 0),\ v = (5, 0, -1, 0)$$

The inequality class of this span is $(\underset{-\infty}{<}, \underset{-\infty}{>}, >, =)$ and, as we can show, it is not an orbit under the action by monomial matrices. Now, the span of these vectors is properly contained by the span of

$$u^* = (-\infty, 5, 1, 0),\ v^* = (5, -\infty, -1, 0)$$

and, for any $a < 5$, every subspan of $\text{span}(u^*, v^*)$ is the span of vectors of the form

$$u' = (a, 5, 1, 0),\ v' = (5, a, -1, 0)$$

We observe that in every vector span offered above, we have $\Delta_3^{u,v} = 1-(-1) = 2$. We know that monomial matrices can only scale and permute the indices of these vectors, hence for every M, we are guaranteed

$$2 \in \Delta^{Mu,Mv}.$$

Which means that given $\text{span}(a, b)$ of inequality class $(\underset{-\infty}{<}, \underset{-\infty}{>}, >, =)$ such that $2 \notin \Delta^{a,b}$, we have that for all monomial matrices M, $span(Mu, Mv) \neq (a, b)$.

With this example, we know that the inequality class with tuple $(\underset{-\infty}{<}, \underset{-\infty}{>}, >, =)$ is not an orbit under the group action. Future work on this project might aim to figure out which of the 13 inequality classes of $G_{tr}(2, 4)$ are well-behaved and which are not. Even better would be discovering some rule for enumerating well-behaved inequality classes in $G_{tr}(2, n)$ for arbitrary n.

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